Type MS · Cloud storage: what it is, pros, cons and security

Answer 1 of 5 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

2(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • (Touch)screen (1) • Speaker (1) • Vibration (motor) (1) • Flashlight/LED light/torch (1) • NFC chip (1)	**Accept:** • display for screen **Do not accept:** • monitor • printer • headset • other peripheral devices	**2**
Question	Answer	Additional Guidance	Mark
2(b)(i)	**The only correct answer is D** *A is not correct because cache is located closer to the CPU* *B is not correct because random-access memory is a physical card on the motherboard* *C is not correct because read-only memory cannot be written to*		**1**
Question	Answer	Additional Guidance	Mark
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Often held/stored remotely/stored on the internet (1) • Held/managed by a third party (1) • Requires internet connectivity to access (1)		**2**

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Cloud storage: what it is, pros, cons and security

Answer 1 of 5 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for any of the following up to a maximum of **three** marks: • Clock speed/number of cycles per second (1) • Number of processor cores (1) • Size of cache (memory) (1) • Type of cache (1) • Width of data bus/word length (1) • Width of address bus (1) • Whether the CPU is overclocked or not (1) • Pipelining architecture (1) • Thermal throttling / cooling system (reducing clock speed) (1)	**Accept:** • units of clock speed e.g. GHz for MP1 • width of bus for either MP5 or MP6 but not both **Do not accept:** cache without clarification of size or type	3
Question	Answer	Additional Guidance	Mark
2(c)(ii)	Award up to **two** marks for a linked description such as: • The CPU places an address on to the address bus (1) to specify the location in main memory (1) • (The address bus) contains the location in RAM (1) to read from/write to (1) • The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1) • The address bus carries the memory location (1) of the current/next instruction (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for any of the following up to a maximum of **two** marks: • Sub-atomic particles (1) • Qubit(s) (1) • Entanglement (1) behaviour of a pair of qubits can influence each other • Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously	**Do not accept:** 'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question	2
		Total for question 2	**12**

Type MS · Cloud storage: what it is, pros, cons and security

Answer 1 of 5 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Cloud storage: what it is, pros, cons and security

Answer 2 of 5 · Question 2 part a · 2 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

2(a)	Award **one** from each: Zafer: 1. Choose strong (1) authentication credentials (username/password) 2. Not sharing his credentials (1) 3. Not having automatic login to cloud / leaving his machine unattended (1) 4. Changing password regularly (1) 5. Password protect documents (1) Provider: 1. Infrastructure (e.g. firewall/servers) must be secure from unauthorised access (1) 2. Keeping their security software up to date (1) 3. Policy and procedures effective in preventing an insider attack / a data breach (1) 4. Data protection laws in the resident country must be obeyed (1) 5. Backup and restore procedures (1) 6. Encryption (1)		2

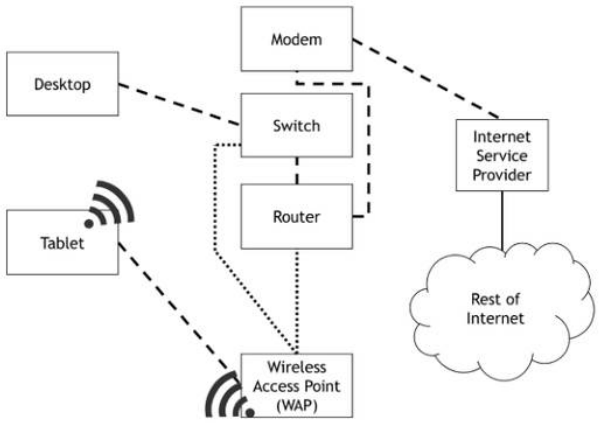
Question Number	Answer	Additional Guidance	Mark
2(b)	**Two** marks if completely correct. **One** mark for any two correct cells. **Robert** / **Zafer** Washing machine design drawings: Read / Read A folder of manuals for new machines: None / Read, Write Or Write, Read	Ignore spelling mistakes	2

Type MS · Cloud storage: what it is, pros, cons and security

Answer 2 of 5 · Question 2 part a · 2 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)	Award **one** mark from each category (maximum of 2) Input: • www.pearson.com / website domain (1) • Uniform Resource Locator / URL (of Pearson website) (1) Output: • 2.20.38.113 (1) • Pearson's machine IP(v4) address (1)	Don't penalise spelling or typos, especially in addresses	2

Question Number	Answer	Additional Guidance	Mark
2(d)	One mark for each dotted connection, correctly located. Maximum of 6 marks. Note: WAP can be connected to either the Switch or the Router, but not both  		6

Type MS · Cloud storage: what it is, pros, cons and security

Answer 2 of 5 · Question 2 part a · 2 marks

2019 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

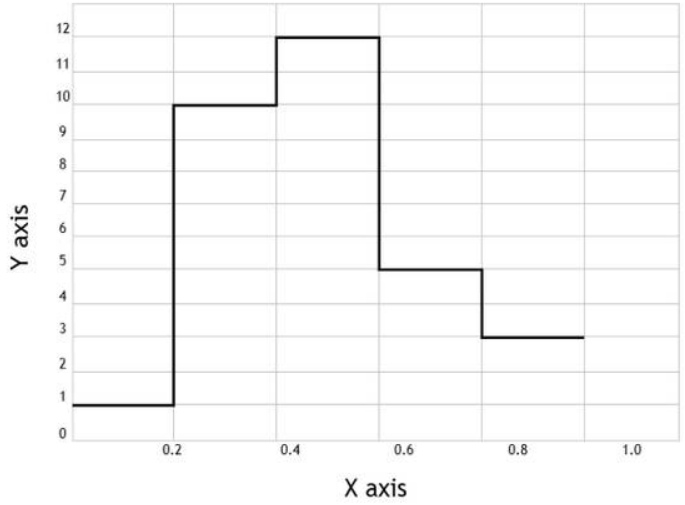
Type MS · Cloud storage: what it is, pros, cons and security

Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table>
 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

Type MS · Cloud storage: what it is, pros, cons and security

Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
Larger file size / less compression (1)
Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
Storage host could be targeted by hackers (1)
Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Cloud storage: what it is, pros, cons and security

Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · Cloud storage: what it is, pros, cons and security

Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark

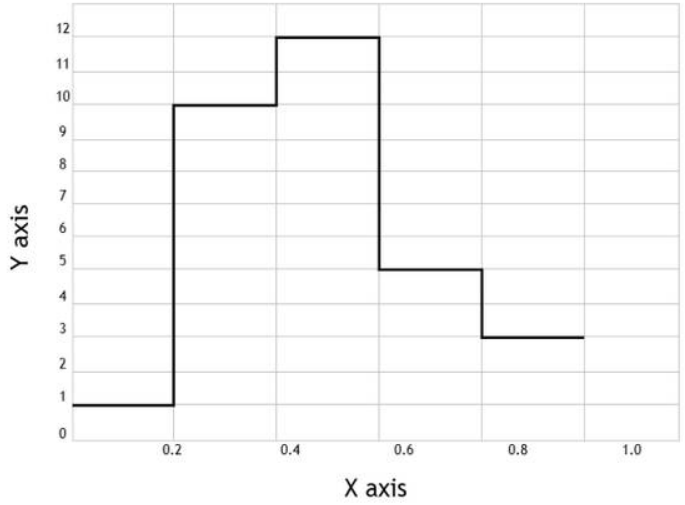
Type MS · Cloud storage: what it is, pros, cons and security

Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table>
 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

Type MS · Cloud storage: what it is, pros, cons and security

Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
Larger file size / less compression (1)
Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
Storage host could be targeted by hackers (1)
Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Cloud storage: what it is, pros, cons and security

Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · Cloud storage: what it is, pros, cons and security

Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark

Type MS · Cloud storage: what it is, pros, cons and security

Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

6(a)	Award **one** mark for each of: Method • Flowchart (1) Justification • It is a visual representation / does not use many words / does not rely on use of English language (1) • It does not rely on understanding specific syntax (1) • Overview without unnecessary detail (1) OR Method • Pseudocode/structured English/written description (1) Justification • It does not rely on understanding specific syntax (1) • It uses English language words (1)	Valid method must be specified first before mark for justification can be awarded.	2

Type MS · Cloud storage: what it is, pros, cons and security

Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for any of the following: • The test on line 3 has already limited the values to 1 and 2 (1) • The only value left for inNum to be (on line 7) is 2 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award up to **two** marks for a linked description, such as: • The oldIndex value will go from highest to lowest (1) the step value is negative/decremented by 1 (each iteration) (1)	Allow 4 to 0 instead of highest and lowest	2

Question Number	Answer	Additional Guidance	Mark
6(c)(ii)	Award **one** mark for any of the following: • Reverse the (array of) scores (1) • Copy the oldScores into newScores, in reverse order (1)		1

Type MS · Cloud storage: what it is, pros, cons and security

Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part d in this mark scheme extract

Question Number	Indicative Content	Additional Guidance	Mark
6(d)	**Growth** • Higher bandwidth available both on wired and wireless networks • Reduced costs of large capacity storage devices • Improved levels of encryption / authentication • Increased trust in service providers • Growth of service provider options • Increased business models that aid scalability regarding storage capacity and costs **Benefits** • Sharing / collaboration on documents and setting of access permissions • Access from any location connected to a WAN • Reduced cost of local computing devices with less storage required • Built in backups of data on server-based systems **Drawbacks** • Increased security risks incurred from outsourcing a service • Relies on high-speed network connections • Reliance on service providers continuing to provide a high level of service • Increased chances of external data breaches • Identifying responsibilities for data breaches / legal jurisdiction • Increased power consumption from server systems and associated cooling systems • Increased use of networks and network traffic which will have an impact on other network users **Conclusion** • A conclusion does not have to be stated explicitly, but can be inferred from the discussion		6

Type MS · Cloud storage: what it is, pros, cons and security

Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Cloud storage: what it is, pros, cons and security

Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part d in this mark scheme extract

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